UMKC Impact & Translation Series

Financial Basics for UMKC Startups

Helping you move your work beyond the university

Financial planning can feel intimidating, especially if your training has focused on research rather than business. But if your work may have potential beyond the university, understanding a few core financial ideas can make a major difference.

Whether your path involves a startup, licensing, external partnership, pilot deployment, or another route to use, financial planning helps you understand what progress will require. It helps you estimate costs, anticipate risks, connect resources to milestones, and make more informed decisions about what path is realistic.

This whitepaper introduces the core financial concepts most relevant to early translation: the three basic financial statements, several key operating metrics, common funding options, and the importance of matching financial planning to the stage and pathway of your work.

UITS Whitepaper

This whitepaper is part of the **UMKC Impact & Translation Series (UITS)** - a set of short videos and supporting materials designed to help faculty think more clearly about how their work can move beyond the university and into real-world use.

The videos introduce core ideas. This document goes deeper on one of those ideas: **how financial planning supports the process of moving research, innovation, and creative work toward real-world impact.**

Where This Fits in the Translation Process

Funding becomes relevant at multiple points in the translation process, but its role changes depending on the stage of the work.

In early stages, funding often supports exploration and validation. This may include feasibility studies, proof-of-concept work, or early customer discovery. At this stage, the goal is not scale. The goal is reducing key uncertainties.

As work progresses, funding begins to support more defined pathways. A licensing strategy, startup pathway, or partnership model may each require different types of capital, timelines, and levels of investment.

Understanding funding in this way shifts the question from “How do I raise money?” to “What does this next step require, and what type of funding actually fits that need?”

Why Financial Planning Matters

In academic work, we are trained to support ideas with evidence. In translation, financial planning is part of that evidence. It shows that you have started thinking not only about the strength of the idea, but also about what it would take to move that idea into use.

That matters whether you are speaking with a potential licensee, an industry collaborator, an internal university partner, a grant reviewer, or an investor. A realistic financial model helps clarify the scale of the opportunity, the likely cost of progress, the timing of key decisions, and the resources required to reach the next milestone.

Knowing how to use financial information to communicate your innovation can make a meaningful difference when working with partners, funders, or collaborators. It helps you connect funding to a specific next step, identify risks before they become constraints, and demonstrate that your project is being approached with discipline rather than optimism alone.

Done well, financial planning does three things:

- **Translates science to business.** Investors can't compare "novel polymer" to "ML platform," but they can compare gross margins and burn.
- **Flags runway gaps early.** A basic cash-flow forecast tells you when to call the Missouri SBDC before panic sets in.
- **Builds credibility.** A tidy income statement suggests you'll keep grant funds tidy, too.

Reality check: The fastest way to lose a deal isn't a flawed patent—it's sloppy numbers.

Meet Your Core Statements

As your work begins moving toward external use, three financial statements become increasingly important: the Income Statement, Balance Sheet, and Cash-Flow Statement.

Think of the Income Statement, Balance Sheet, and Cash-Flow Statement as a trio of UMKC basketball stats. One tells you the score, one the roster, and one how gassed the players are.

Statement	What It Answers	Common Rookie Mistake
Income Statement (P&L)	Are we making or losing money?	Ignoring customer-support or hosting costs, which overstates margins.
Balance Sheet	What do we own and owe today?	Celebrating bookings while receivables age to 90+ days.
Cash-Flow Statement	Do we have enough gas in the tank to reach the next milestone?	Focusing on "profit" while burn accelerates.

UMKC Tip: *Keep these linked in one spreadsheet so every assumption ripples automatically.*

Metrics That Help You Evaluate Viability

Most external reviewers will not focus equally on every line item. They often look first at a small set of indicators that suggest whether the model is economically plausible, whether costs are understood, and whether growth would become more sustainable over time.

Metric	Why It Matters	Benchmarks & Watch-outs
Gross Margin	Shows pricing power and operating leverage.	70–90 % for mature SaaS; hardware trends lower.
CAC (Customer Acquisition Cost)	Measures how much it costs to land a customer.	Must fall over time; if CAC > LTV, you're upside-down.
LTV (Lifetime Value)	Total revenue per customer minus support costs.	Aim for LTV ≥ 3× CAC.
Burn Rate & Runway	Cash out per month and how long you last.	Keep ≥ 12 months runway pre-Series A.

These metrics do not tell the whole story, but they can quickly reveal whether the path you are modeling is becoming more or less viable.

Financial Implications of Different Go-to-Market Paths

One of the most important translation decisions is identifying who will adopt, buy, implement, or deploy what you are creating. In some cases, that means selling to organizations. In others, it may involve direct consumer adoption. In still others, the path may involve licensing or institutional implementation rather than immediate sales.

The financial structure of these paths can look very different. Customer acquisition, onboarding, support, and payment timing all affect what kind of model is realistic. Thinking through those differences early can prevent teams from choosing a path that sounds attractive but is financially difficult to sustain.

Whether you're pitching to hospital systems or TikTok users, success boils down to four cost buckets: acquisition, onboarding, collections, and support. Each behaves differently in B2B and B2C terrain. Model them honestly and you'll know which go-to-market path your innovation can actually afford.

Cost Bucket	B2B Flavor	B2C Flavor	Keep an Eye On
Acquisition	Long sales cycle; trade shows, pilots.	Digital ads, viral loops; quick but pricey at scale.	Track CAC by channel every quarter.

Onboarding	Custom integrations, staff training.	Self-serve UX, tutorials.	Capitalize eligible setup costs under GAAP.
Collections	Net-30/60 terms; watch A/R aging.	Instant card capture; higher fees.	Model days-sales-outstanding.
Support & Retention	Account managers, lower churn.	High ticket volume, higher churn.	Treat support headcount as COGS to keep margins honest.

Bottom line: The key question is not whether B2B or B2C sounds more exciting. The key question is whether the economics of that path are realistic for your stage, team, and intended users.

Funding the Next Meaningful Step

Funding is often discussed as if it were the goal. In practice, funding is useful only if it helps a project reach a meaningful next milestone. That milestone might be technical validation, regulatory planning, customer discovery, prototype refinement, pilot deployment, or some other step that reduces uncertainty and improves decision-making.

Different kinds of funding support different kinds of progress. Some sources preserve ownership but come with constraints. Others offer speed or flexibility but change control and long-term economics. The right funding approach depends on what you are trying to accomplish next, how much risk has already been reduced, and which pathway to impact appears most realistic.

A useful question to ask is: What is the next concrete milestone, and what kind of capital best fits that milestone?

Debt vs Equity in Plain English

- **Debt**: Cheap control, tax-deductible interest, but you need collateral and cash flow.
- **Equity**: Easier to get for pre-revenue tech, but every share you sell is a share you no longer own.

Non-dilutive & Creative Capital

Mix and match:

- **SBIR/STTR grants** for tech milestones.
- **Pitch contests** (UMKC's Roo Idea events) for quick cash and visibility.
- **Convertible notes / SAFEs** to bridge between grants and priced rounds.

Map each chunk of money to a milestone *before* you cash the check.

Building a Model You Can Defend

Models aren't crystal balls; they're hypothesis-testing labs. The real value is seeing what happens when you nudge a dial—raise price, delay a grant, or spike server costs.

Guiding Question:
What would it actually take to move my work toward real-world use, and how can financial planning help me understand the resources, risks, and next step required to get there?

Build an integrated 3-statement model that updates automatically so you're never caught off-guard.

1. **Start with hard data** - market sizes, conversion rates, supplier quotes. Save each source in a "data room."
2. **Mix top-down and bottom-up forecasting** so ambition meets reality.
3. **Add supporting schedules** for working capital: receivables, inventory, payables.
4. **Run at least three scenarios:** base, downside, aggressive and know what triggers each.

If you can't explain an input in 30 seconds, it doesn't belong in the model.

Avoiding the Faceplants

Every month we watch promising projects trip on the same banana peels—over-optimistic sales curves, forgotten support costs, or a cash crunch hiding behind healthy revenue.

This quick-hit checklist will save you painful investor meetings:

Pitfall	Quick Fix
Ignoring indirect costs	Add cloud hosting, compliance, and support to COGS.
Over-optimistic revenue	Cross-check against market size *and* sales capacity.
Under-staffing forecasts	Use revenue-per-employee benchmarks as a sanity check.
Skipping working-capital timing	Build a 12-month operational cash-flow grid.
Sparse communication	Send a one-page finance dashboard to stakeholders monthly.

Action Steps (Your 30-Day Plan)

If this paper raised useful questions about your work, here are five practical next steps:

1. Build a simple pro forma financial model for the next stage of your project, even if the numbers are rough.

2. Identify the next meaningful milestone and estimate the time, cost, and people required to reach it.
3. Stress-test your assumptions about pricing, adoption, support costs, and timing.
4. Explore what kind of funding best fits your current stage, including non-dilutive options where appropriate.
5. Use that financial picture to clarify whether your work is better suited for licensing, partnership, startup formation, or further validation before choosing a path.

Do these and you'll be ahead of 90% of academic startups.

Conclusion

Strong research does not automatically become real-world impact. Moving work beyond the university requires choices about pathway, timing, resources, and risk. Financial planning helps make those choices clearer.

You do not need a perfect model at the outset. But you do need enough clarity to understand what the next step would require, what assumptions you are making, and whether the path you are considering is realistic.

Used well, financial planning is not just a business exercise. It is part of translation. It helps you move from a promising idea to a more grounded understanding of what it would take for that work to create impact beyond the university.

Resources & Contacts

UMKC Office of Commercialization
Director: Jeff Blackwood, MBA
📍 Website: www.umkcinnovates.com/tech-commercialization
📧 Email: techcomm@umkc.edu
📞 Phone: 816-235-6373

- **Missouri SBDC @ UMKC** – bookkeeping & cash-flow workshops
- **Propel Smeal Financial Guides** – free templates & tutorials
- **EY Finance Navigator (academic license)** – 3-statement modeling
- **KCSourceLink** – regional funding database and mentor network

(We're here to help—reach out early and often.)